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## Governed Bilingual Inputs

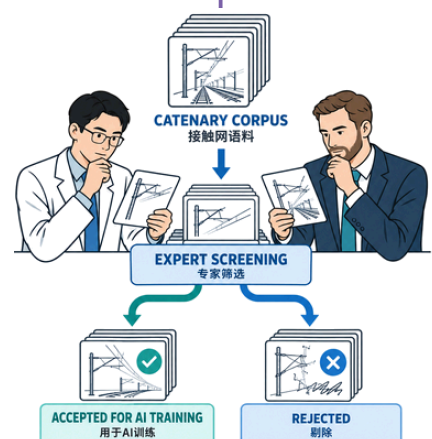


$q_{zh}$ : 接触网供电的额定电压是多少

parallel inputs

$q_{en}$ : What is the rated voltage for contact network power supply

**Governed bilingual records**  
Chinese/English · source review · split



**Admissibility filter**  
approved / test excluded

Query-time approval only; not a trainable mask.

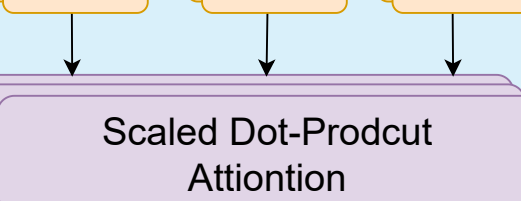
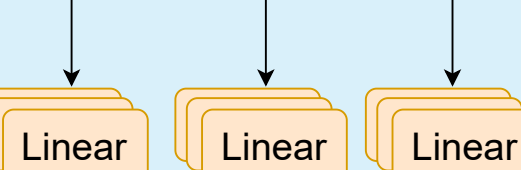
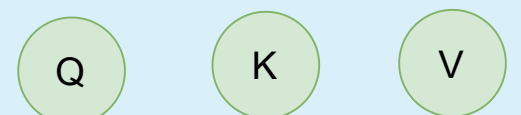
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## Shared Neural Encoding

**Frozen shared BGE-M3 encoder**  
shared query/evidence parameters

Token + positional embeddings

### Multi-Head Self-Attention



Concat

Linear

Residual + Add & Norm

### Feed-Forward Network

Residual + Add & Norm x N

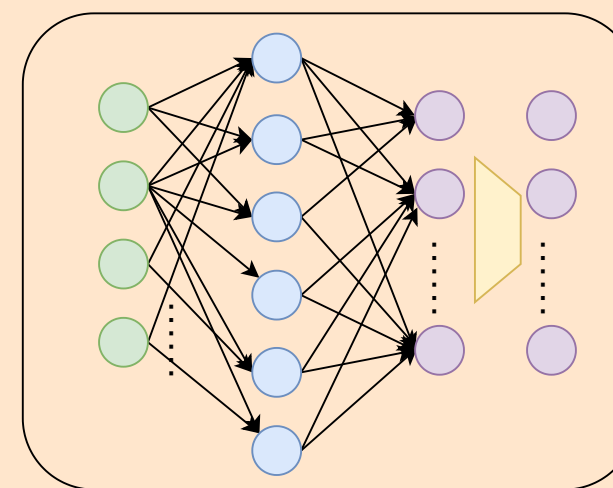
$$z_q = \text{Encoder}_{\text{BGE-M3}}(q)$$

$$z_d = \text{Encoder}_{\text{BGE-M3}}(d)$$

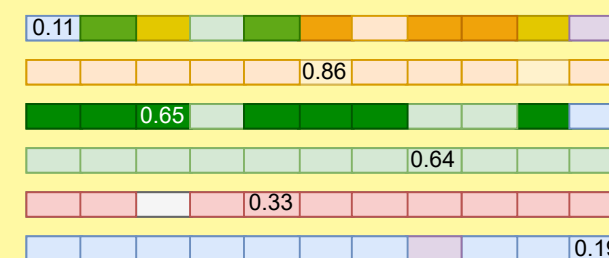
$$z_q, z_d \in \mathbb{R}^{1024}$$

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## Parallel Hybrid Retrieval



### Embedding Matrix



**Dense retrieval**  
 $\cos(z_q, z_d)$   
vector rank

**BM25 retrieval**  
lexical rank

**Reciprocal Rank Fusion**  
 $\text{RRF}(d) = \sum 1/(k_0 + \text{rank})$

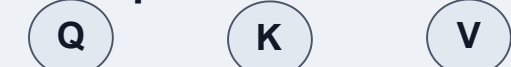
**Top-k evidence passages**  
[E1]/[E2] passage + source

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## Evidence-Conditioned LLM + QLoRA

**Serialized context**  
[Instr] | [Q] | evidence | [A]

### Repeated decoder



**Masked causal attention**

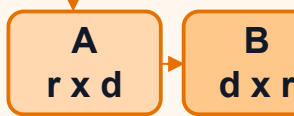
Residual + Add & Norm

**Gated FFN**

Residual + Norm x N

state

**QLoRA**  
low-rank update



$$dW = BA$$

$$r = 64; \alpha = 16$$

**One shared LM head + Softmax**  
 $p(y_t | X, y_{<t})$

**Autoregressive bilingual answer**  
token1 -> token2 -> token3 -> ...  
evidence labels: [E1] / [证据1]

**Independent evaluation**  
external metrics / no gradient

**Railway QA**  
F1 / containment

**Citations**  
coverage / syntax

**Translation**  
BLEU / chrF / COMET

**Evidence support**  
semantic proxy

**Resource-constrained inference**  
latency / GPU memory / tokens/s